

	compared to weight of wire	A1	[4]
7	(a) 'uniform' distribution	B1	[1]
	(b) concentric rings	B1	[1]
	(c) higher speed, more momentum	M1	
	$\lambda = h / p$	M1	
	so λ decreases and ring diameter decreases	A1	[3]
8	(a) arrow labelled E pointing down the page	B1	[1]
	(b) (i) $Bqv = qE$	M1	
	forces are independent of mass and charge 'cancels'	M1	
	so no deviation	A1	[3]
	(ii) magnetic force > electric force	M1	
	so deflects	M1	
	'downwards'	A1	[3]